

ECE 332: Engineering Electromagnetics II

Lecture 14

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Quick Recap

Waveguides

A **waveguide** is a **higher-order transmission lines** that carries EM waves where one of $\mathbf{E}$ or $\mathbf{H}$ is orthogonal to the direction of propagation $\hat{\mathbf{k}}$ but not both.

An **optical fiber** is another example of this type of transmission line. **Metal waveguides** use reflection at the boundary between a dielectric medium and a conductor to direct EM waves.

A **mode** is

- **transverse electric (TE)** if $\mathbf{E}$ is orthogonal to $\hat{\mathbf{k}}$ but not $\mathbf{H}$,
- **transverse magnetic (TM)** if $\mathbf{H}$ is orthogonal to $\hat{\mathbf{k}}$ but not $\mathbf{E}$.

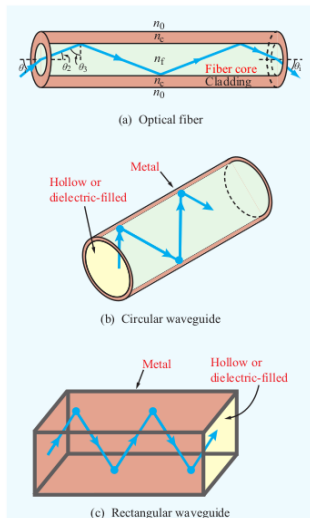


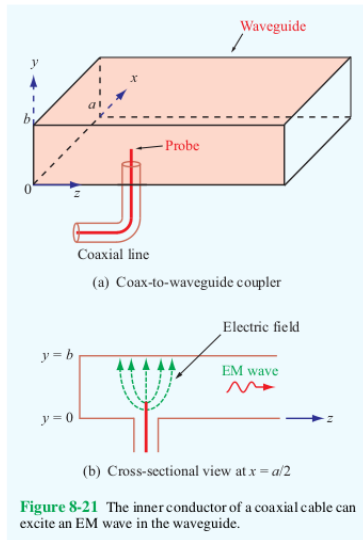
Figure 8-20 Wave travel by successive reflections in (a) an optical fiber, (b) a circular metal waveguide, and (c) a rectangular metal waveguide.

Quick Recap (Continued)

Signals can be **transferred between coaxial cables and metal waveguides** as shown in the figure. **Coaxial cables** are appropriate for transmitting frequencies **below 30 GHz** but **metal waveguides** are a good alternative for **high-power RF frequencies from 5 to 100 GHz**.

Rectangular Metal Waveguides

We explored the **modes**, i.e. types of EM waves, that will propagate through a **rectangular metal waveguide** that is filled with a **lossless dielectric material**, which is often air. The waveguide has **width a** and **height b** , where $a > b$.

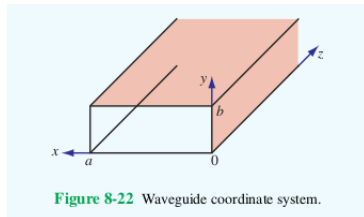


Quick Recap (Continued)

The **Cartesian coordinate system** shown in the adjacent figure is appropriate for this purpose, where the **direction of propagation** of the EM waves is $\hat{z}$. In this coordinate system, the phasors of the electric and magnetic fields have the form

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{x} + \tilde{E}_y \hat{y} + \tilde{E}_z \hat{z} \quad (1)$$

$$\tilde{\mathbf{H}} = \tilde{H}_x \hat{x} + \tilde{H}_y \hat{y} + \tilde{H}_z \hat{z}. \quad (2)$$



The modes can be divided into

- **transverse magnetic modes** with $\tilde{\mathbf{H}}$ orthogonal to $\hat{z}$, i.e. $\tilde{H}_z = 0$,
- **transverse electric modes** with $\tilde{\mathbf{E}}$ orthogonal to $\hat{z}$, i.e. $\tilde{E}_z = 0$.

Only EM waves in these modes propagate through the waveguide.

Quick Recap (Continued)

Assuming that the waves propagate in the $+z$ direction, using Maxwell's equations, and imposing appropriate boundary conditions, the m th **transverse magnetic mode** $\mathbf{TM}_{mn}$ has components

$$\tilde{E}_x(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (3)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (4)$$

$$\tilde{E}_z(x, y, z) = E_0 \sin \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (5)$$

$$\tilde{H}_x(x, y, z) = \frac{j\omega\epsilon}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (6)$$

$$\tilde{H}_y(x, y, z) = -\frac{j\omega\epsilon}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (7)$$

$$\tilde{H}_z(x, y, z) = 0, \quad (8)$$

where $m = 1, 2, \dots$ and $n = 1, 2, \dots$,

Quick Recap (Continued)

and the mn^{th} **transverse electric mode** $\mathbf{TE}_{mn}$ has components

$$\tilde{E}_x(x, y, z) = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (9)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (10)$$

$$\tilde{E}_z(x, y, z) = 0 \quad (11)$$

$$\tilde{H}_x(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (12)$$

$$\tilde{H}_y(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (13)$$

$$\tilde{H}_z(x, y, z) = H_0 \cos \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z}, \quad (14)$$

where $m = 1, 2, \dots$ and $n = 0, 1, \dots$, assuming that $a > b$.

Quick Recap (Continued)

Here, E_0 and H_0 are **amplitudes**, the **cutoff wavenumber** is

$$k_c^2 = k^2 - \beta^2 = \omega^2 \mu \epsilon - \beta^2, \quad (15)$$

the **unbounded-medium wavenumber** is

$$k = \omega \sqrt{\mu \epsilon}, \quad (16)$$

and the **phase constant** is

$$\beta = \sqrt{\omega^2 \mu \epsilon - (m\pi/a)^2 - (n\pi/b)^2}. \quad (17)$$

For each mode, with specific values for m and n , the lowest frequency for a wave that will propagate through the waveguide is the **cutoff frequency**

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (18)$$

where

$$u_{p0} = 1/\sqrt{\mu \epsilon} \quad (19)$$

is the **speed of propagation in an unbounded medium**.

Quick Recap (Continued)

The phase constant can be expressed in terms of the cutoff frequency

$$\beta = \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \quad (20)$$

and the **phase velocity** or **speed of propagation** of a TM or TE wave in the waveguide is

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad (21)$$

These both **depend on the frequency** f of the EM wave.

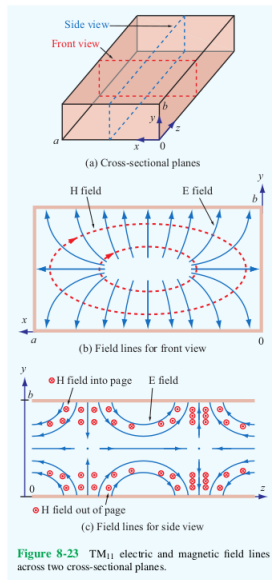
The **wave impedance** in the waveguide is

$$Z_{\text{TM}} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \quad (\text{TM modes}) \quad (22)$$

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}} \quad (\text{TE modes}). \quad (23)$$

Quick Recap (Continued)

The **dominant mode** is the mode with the **lowest cutoff frequency**. This is TM_{11} for the TM modes and TE_{10} for the TE modes. The adjacent figure shows the fields of the mode TM_{11} .



Quick Recap (Continued)

Table 8-3 Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$, filled with a dielectric material with constitutive parameters ϵ and μ . The TEM case, shown for reference, pertains to plane-wave propagation in an unbounded medium.

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/Z_{TE}$ $\tilde{H}_y = \tilde{E}_x/Z_{TE}$ $\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y/Z_{TM}$ $\tilde{H}_y = \tilde{E}_x/Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = E_{x0} e^{-j\beta z}$ $\tilde{E}_y = E_{y0} e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/\eta$ $\tilde{H}_y = \tilde{E}_x/\eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $\beta = k \sqrt{1 - (f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0} / \sqrt{1 - (f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega \sqrt{\mu\epsilon}$ $u_{p0} = 1/\sqrt{\mu\epsilon}$

Quick Recap (Continued)

Propagation Velocities

Information is **encoded** into EM waves by modifying their **amplitude**, **frequency**, or **phase**. A modified wave has an **envelope**, which is the shape encapsulating the wave, and consists of the **frequency components** associated with its Fourier representation. The envelope propagates at the **group velocity**

$$u_g = 1/(d\beta/d\omega). \quad (24)$$

For **TM** or **TE** mode mn in a **rectangular waveguide**,

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \quad (25)$$

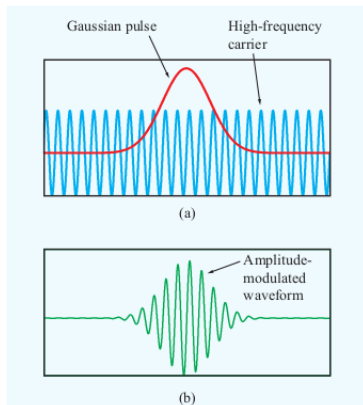


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Quick Recap (Continued)

and the individual frequency components propagate at the **phase velocity**

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad (26)$$

Here,

$$u_p u_g = u_{p0}^2 \quad (27)$$

and, for the **waves that propagate through the waveguide**, $f > f_{mn}$. So,

- $u_p > u_{p0}$ and $u_g < u_{p0}$
- $\lim_{f \rightarrow \infty} u_p = \lim_{f \rightarrow \infty} u_g = u_{p0}$.

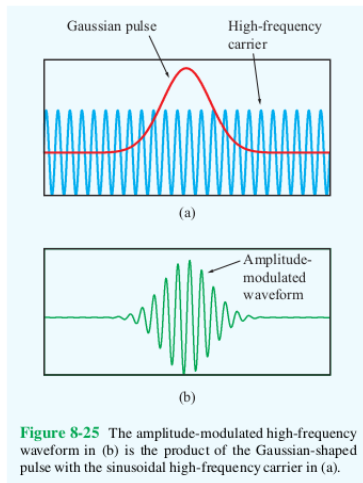


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Quick Recap (Continued)

An ω - β **diagram** shows **how TM or TE modes propagate** in a waveguide compared to **how they would propagate in an unbounded medium**. At a point on a curve in this diagram,

- $u_p = \omega/\beta$,
- $u_g = d\omega/d\beta$ is the slope of the curve.

A curve for mode mn has slope 0 and $\omega = 2\pi f_{mn}$ at $\beta = 0$. So, at this point, $u_g = 0$ and $u_p \rightarrow \infty$ as $\beta \rightarrow 0^+$.

The **TEM line** shows ω vs β when the medium is **unbounded**. On this line, $u_g = u_p = u_{p0}$ and the curves for all modes approach the TEM line as $f \rightarrow \infty$.

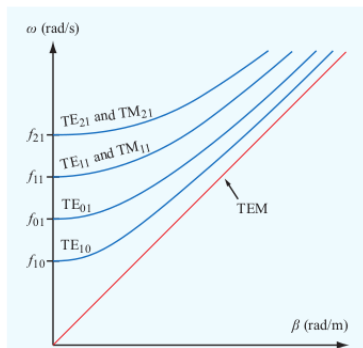


Figure 8-26 ω - β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

Quick Recap (Continued)

The above results pertain to **straight-line travel through the medium** in a waveguide and don't consider the consequences of reflection. Determining the **overall propagation velocity** requires accounting for the **zigzagging** of the path that a wave follows through the waveguide.

Quick Recap (Continued)

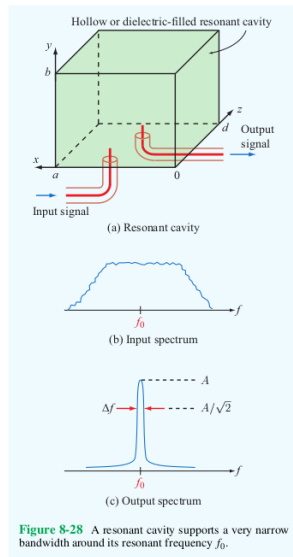
Cavity Resonators

A **cavity** is a waveguide where the open ends have been replaced with metal walls. For the $a \times b \times d$ rectangular cavity shown in the figure with given the coordinate system, the boundary conditions imply

$$\beta = \frac{p\pi}{d}, \quad p = 1, 2, \dots \quad (28)$$

A mode in the cavity is specified by the integers (m, n, p) . The cavity only sustains a mode at the **resonant frequency**

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (29)$$



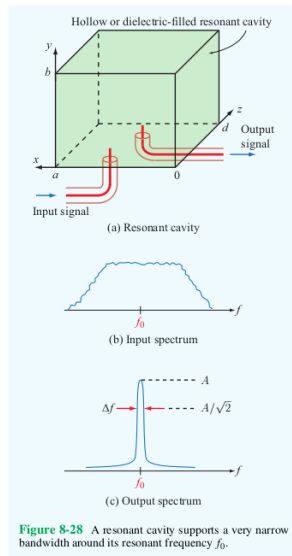
Quick Recap (Continued)

For the TM modes, m and n start at 1 and p starts at 0. For the TE modes, m and n start at 0 and p starts at 1.

Although, in the ideal case, only the wave with frequency f_{mnp} persists in the cavity, a physical cavity will sustain waves with a small range of frequencies centered at f_{mnp} . A measure of this is the **quality factor**

$$Q \approx \frac{f_{mnp}}{\Delta f}, \quad (30)$$

where Δf is measured at the point on the peak that is $f_{mnp}/\sqrt{2}$. Here, $\frac{\Delta f}{f_{mnp}}$ is the **normalized bandwidth** of the cavity.



Chapter 9: Radiation and Antennas

An **antenna**

- converts a guided EM wave into a wave that propagates in an unbounded medium, i.e. **transmits** the EM wave into the unbounded medium,
- or converts an EM wave propagating in an unbounded medium into a guided EM wave, i.e. **receives** the EM wave from the unbounded medium.

The **size** of an antenna is measured in units of the **wavelength** λ it is intended to transmit or receive.

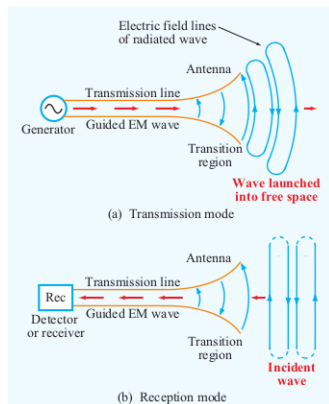
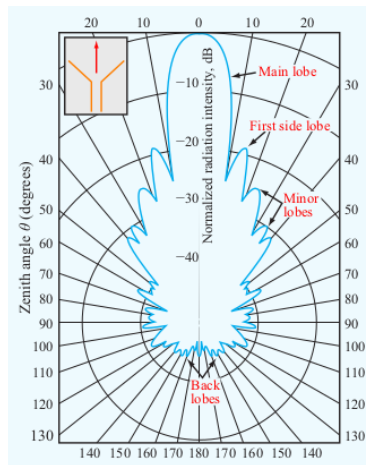


Figure 9-1 Antenna as a transducer between a guided electromagnetic wave and a free-space wave, for both transmission and reception.

Radiation and Antennas (Continued)

The **antenna radiation pattern** or **antenna pattern** of an antenna is the **spatial distribution of relative power** that the antenna radiates when it transmits. The adjacent figure shows an example. For most antennas, the **relative sensitivity as a receiver** to incoming EM waves from all directions **matches its antenna pattern**. This is called **reciprocity** and all antennas considered will be assumed to be reciprocal.

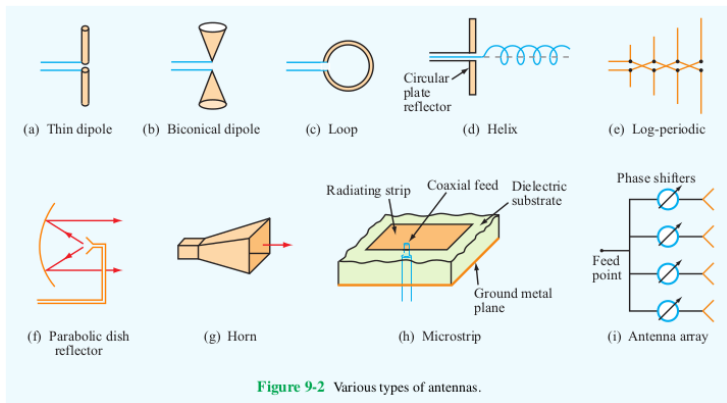
A hypothetical antenna that radiates (or receives) **uniformly in all directions** is called an **isotropic antenna** and is used as a reference.



Radiation and Antennas (Continued)

An antenna is also characterized by the

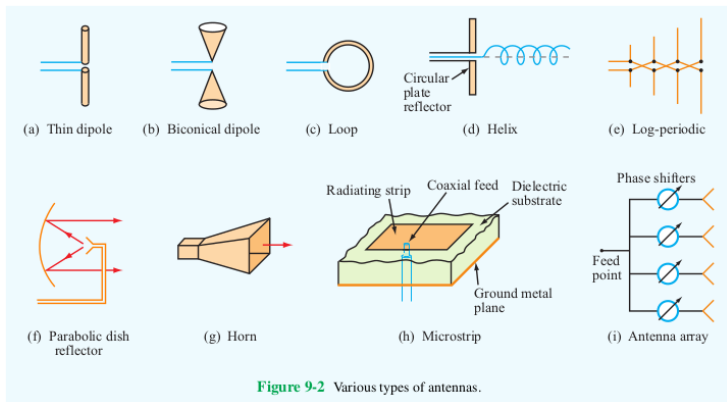
- **antenna polarization**, which is the polarization of the emitted wave when the antenna transmits
- **antenna impedance**, which is associated with the transfer of power from the wave generator to the antenna.



Radiation and Antennas (Continued)

Antennas utilize two types of **radiation sources** to transmit EM waves into unbounded media:

- **current sources**, where time-varying currents produce the EM waves
- **aperture fields**, where the EM fields are generated across an opening/aperture



Radiation and Antennas (Continued)

This distinction is slightly misleading since the ultimate source in both cases is time-varying currents.

Antennas can be combined in an **antenna array** that acts like a single antenna. The **antenna pattern of an array** can be altered to produce a **specific shape** by controlling the waves emitted by each antenna.

As discussed earlier, **point sources** of EM waves **emit spherical waves**. But these waves are **essentially planar** for a receiver that is **sufficiently far from the source**.

If the waves arriving at a receiver are planar, the receiver is said to be in the **far field** of the source. The mathematics of planar waves are much simpler and it is easier to understand the effects of waves in the far field.

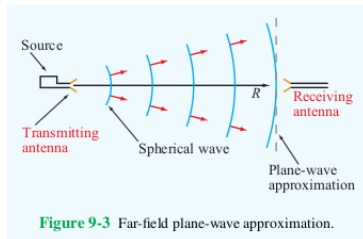


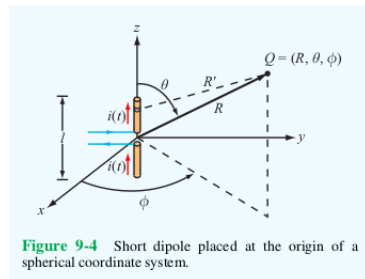
Figure 9-3 Far-field plane-wave approximation.

Hertzian Dipole

A **Hertzian dipole** is a **thin, linear conductor** with length ℓ that is very short compared to the wavelength λ of the EM waves it is intended to produce.

The textbook specifies

$$\ell \leq \lambda/50. \quad (31)$$



The current in the wire will be assumed to be the same along the entire wire and to have the form

$$i(t) = I_0 \cos \omega t \text{ (A)}, \quad (32)$$

where I_0 is the amplitude. Since $i(t) = \text{Re} \{ I_0 e^{j\omega t} \}$, the phasor of $i(t)$ is

$$\tilde{I} = I_0. \quad (33)$$

Hertzian Dipole (Continued)

The EM fields at a point Q with position vector $\mathbf{R}$ can be determined using the **retarded vector potential $\mathbf{A}$** . The phasor form of $\mathbf{A}$ at $\mathbf{R}$ in a **lossless, nonmagnetic medium**, like air, is

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu_0}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}} e^{-jkR'}}{R'} d\mathcal{V}'. \quad (34)$$

Assuming $\epsilon_r = 1$, the wavenumber of the medium is

$$k = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c} = \frac{2\pi}{\lambda}. \quad (35)$$

For the dipole, $\tilde{\mathbf{J}} = (I_0/s) \hat{\mathbf{z}}$, $d\mathcal{V}' = s dz$, and it's reasonable to assume $R' \approx R$.

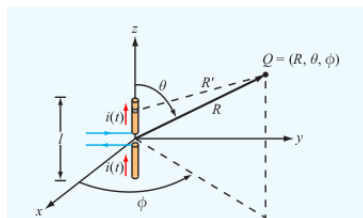


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

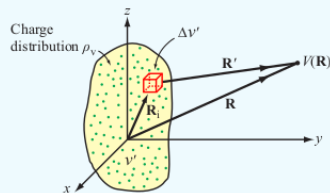


Figure 6-16 Electric potential $V(\mathbf{R})$ due to a charge distribution ρ_v over a volume $\mathcal{V}'$.

Hertzian Dipole (Continued)

So,

$$\begin{aligned}\tilde{\mathbf{A}}(\mathbf{R}) &= \frac{\mu_0}{4\pi} \frac{e^{-jkR}}{R} \int_{-\ell/2}^{\ell/2} I_0 \hat{\mathbf{z}} dz \\ &= \frac{\mu_0 I_0 \ell}{4\pi} \frac{e^{-jkR}}{R} \hat{\mathbf{z}}.\end{aligned}\quad (36)$$

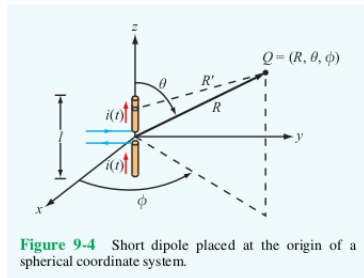
The **spherical propagation factor** is

$$\frac{e^{-jkR}}{R}.\quad (37)$$

It contains all of the R -dependence of $\mathbf{A}(\mathbf{R})$.

Given an expression for $\mathbf{A}$, it is now possible to calculate the phasors of the magnetic and electric field intensities using the equations

$$\tilde{\mathbf{H}} = \frac{1}{\mu_0} \nabla \times \tilde{\mathbf{A}}, \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon_0} \nabla \times \tilde{\mathbf{H}}.\quad (38)$$



Hertzian Dipole (Continued)

These calculations are best done in spherical coordinates. The components of $\mathbf{A}$ are

$$\tilde{A}_R = \frac{\mu_0 I_0 \ell}{4\pi} \frac{e^{-jkR}}{R} \cos \theta \quad (39)$$

$$\tilde{A}_\theta = -\frac{\mu_0 I_0 \ell}{4\pi} \frac{e^{-jkR}}{R} \sin \theta \quad (40)$$

$$\tilde{A}_\phi = 0. \quad (41)$$

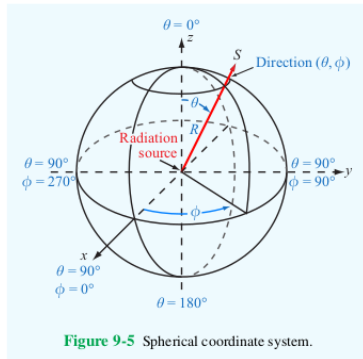


Figure 9-5 Spherical coordinate system.

What does $\tilde{A}_\phi = 0$ mean? It means that the potential is the same regardless of the azimuthal direction.

Hertzian Dipole (Continued)

The components of the electric and magnetic field phasors are

$$\tilde{E}_R = \frac{2I_0\ell k^2}{4\pi}\eta_0 e^{-jkR} \left(\frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right) \cos\theta \quad (42)$$

$$\tilde{E}_\theta = \frac{I_0\ell k^2}{4\pi}\eta_0 e^{-jkR} \left(\frac{j}{kR} + \frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right) \sin\theta \quad (43)$$

$$\tilde{E}_\phi = 0 \quad (44)$$

and

$$\tilde{H}_R = 0 \quad (45)$$

$$\tilde{H}_\theta = 0 \quad (46)$$

$$\tilde{H}_\phi = \frac{I_0\ell k^2}{4\pi} e^{-jkR} \left(\frac{j}{kR} + \frac{1}{(kR)^2} \right) \sin\theta, \quad (47)$$

where $\eta_0 = \sqrt{\mu_0/\epsilon_0}$ is the intrinsic impedance of free space.

Hertzian Dipole (Continued)

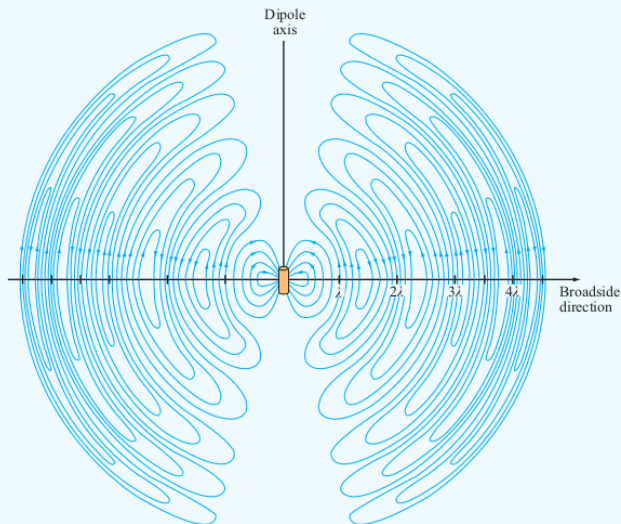


Figure 9-6 Electric field lines surrounding an oscillating dipole at a given instant.

Hertzian Dipole (Continued)

In the **far field**, $R \gg \lambda$. This means

$$kR = \frac{2\pi R}{\lambda} \gg 1 \quad (48)$$

and the terms with $(kR)^2$ and $(kR)^3$ in the denominator can be neglected. So, the components of the electric and magnetic field phasors become

$$\tilde{E}_R = \tilde{E}_\phi = 0 \quad (49)$$

$$\tilde{E}_\theta = \frac{jI_0 \ell k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \text{ (V/m)} \quad (50)$$

and

$$\tilde{H}_R = \tilde{H}_\theta = 0 \quad (51)$$

$$\tilde{H}_\phi = \frac{jI_0 \ell k}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta = \frac{\tilde{E}_\theta}{\eta_0} \text{ (A/m)}, \quad (52)$$

i.e. the components of a TEM plane wave propagating in the $\hat{\mathbf{R}}$ direction.

Hertzian Dipole (Continued)

The **power density** of the EM wave is the **average Poynting vector**

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re} \left\{ \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right\} = S(R, \theta) \hat{\mathbf{R}} \quad (\text{W/m}^2), \quad (53)$$

where

$$S(R, \theta) = \left(\frac{\eta_0 k^2 I_0^2 \ell^2}{32\pi^2 R^2} \right) \sin^2 \theta \quad (\text{W/m}^2). \quad (54)$$

The factor

$$S_0 = \frac{\eta_0 k^2 I_0^2 \ell^2}{32\pi^2 R^2} \quad (55)$$

only depends on R and

$$S(R, \theta) = S_0 \sin^2 \theta \quad (\text{W/m}^2). \quad (56)$$

The **normalized radiation intensity**

$$F(\theta, \phi) = \frac{S(R, \theta, \phi)}{S_{\text{max}}} \quad (57)$$

determines the antenna pattern.

Hertzian Dipole (Continued)

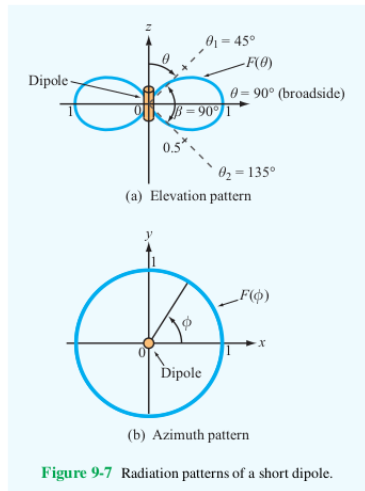
Here, $S(R, \theta, \phi)$ is the power density at a given distance R and $S_{\max}$ is the maximum value of S at R .

For the Hertzian dipole, the maximum power density for a given distance R is when $\sin \theta = 1$, i.e. $\theta = \pi/2$. So,

$$\begin{aligned} S_{\max} = S_0 &= \frac{\eta_0 k^2 I_0^2 \ell^2}{32\pi^2 R^2} = \frac{120\pi (2\pi/\lambda)^2 I_0^2 \ell^2}{32\pi^2 R^2} \\ &= \frac{15\pi I_0^2}{R^2} \left(\frac{\ell}{\lambda}\right)^2 \quad (\text{W/m}^2) \end{aligned} \quad (58)$$

Therefore, the **normalized radiation intensity** is

$$F(\theta, \phi) = F(\theta) = \sin^2 \theta. \quad (59)$$



Antenna Radiation Characteristics

In general, an **antenna pattern** represents the **far-field directional properties** of an antenna at a given distance. It's the average power that the antenna radiates through a sphere centered on the antenna in the far-field.

The **element of area** on the sphere of radius R is

$$dA = R^2 \sin \theta d\theta d\phi = R^2 d\Omega, \quad (60)$$

where

$$d\Omega = \sin \theta d\theta d\phi \text{ (sr)} \quad (61)$$

is the **solid angle** associated with dA .

The unit of solid angle is **steradians**. Since the area of a sphere of radius R is $4\pi R^2$, the surface of a sphere is evenly divided into 4π steradians.

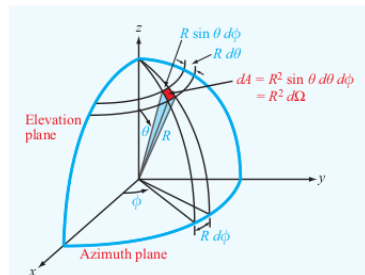


Figure 9-8 Definition of solid angle $d\Omega = \sin \theta d\theta d\phi$.

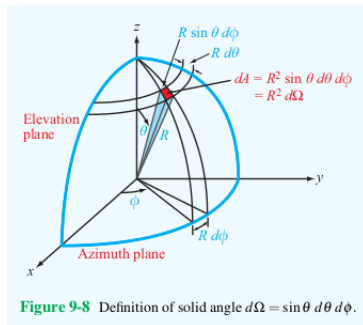
Antenna Radiation Characteristics (Continued)

The power flowing through the area $d\mathbf{A} = dA \hat{\mathbf{R}}$ that is emitted by an antenna at the center of the sphere is

$$\begin{aligned} dP_{\text{rad}} &= \mathbf{S}_{\text{av}} \cdot d\mathbf{A} = \mathbf{S}_{\text{av}} \cdot dA \hat{\mathbf{R}} \\ &= S(R, \theta, \phi) dA \\ &= R^2 S(R, \theta, \phi) d\Omega. \end{aligned} \quad (62)$$

Therefore, the power radiated through the entire sphere is the **total radiated power**

$$\begin{aligned} P_{\text{rad}} &= \int_S dP_{\text{rad}} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} R^2 S(R, \theta, \phi) \sin \theta d\theta d\phi \\ &= R^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} S_{\text{max}} F(\theta, \phi) \sin \theta d\theta d\phi \\ &= R^2 S_{\text{max}} \iint_{4\pi} F(\theta, \phi) d\Omega. \end{aligned} \quad (63)$$



Antenna Radiation Characteristics (Continued)

Antenna Pattern

The antenna pattern, which is encapsulated in the normalized radiation intensity $F(\theta, \phi)$, is a two-dimensional surface in three-dimensional space.

This can be depicted using a

three-dimensional plot.

But the antenna pattern is often represented using two particular two-dimensional cross-sections – the pattern in the **elevation plane**, where ϕ is constant, and the pattern in the **azimuth plane**, where θ is constant. These planes are called the **principal planes** of the spherical coordinate system.

The antenna pattern is usually plotted using a **decibel scale** for the normalized radiation intensity,

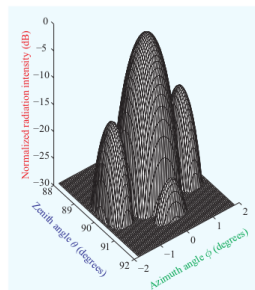


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.

Antenna Radiation Characteristics (Continued)

i.e.

$$F \text{ (dB)} = 10 \log F. \quad (64)$$

Notice that F already incorporates the reference value $S_{\max}$.

The elevation plane is also called the θ plane and the azimuth plane is also called the ϕ plane. These planes can also be plotted using different two-dimensional coordinate systems.

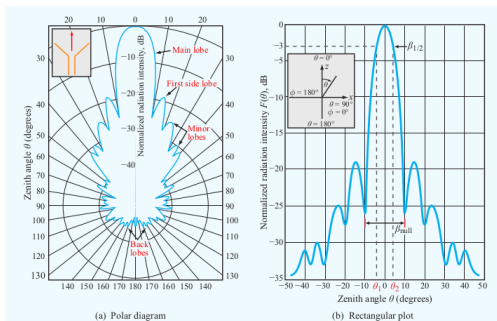


Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

Antenna Radiation Characteristics (Continued)

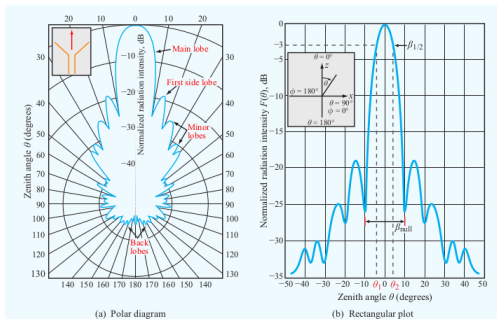


Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

If most of the energy is radiated in a particular direction in one of these plots, the portion of the pattern associated with that direction is called the **main lobe**. Other important but less prominent portions of the pattern are called **side lobes** and peaks that point away from the main lobe are called **back lobes**.

Antenna Radiation Characteristics (Continued)

Beam Dimensions

Another quantity that characterizes an antenna is its **pattern solid angle**. For an antenna with a single main lobe, the pattern solid angle

$$\Omega_p = \iint_{4\pi} F(\theta, \phi) d\Omega \quad (\text{sr}) \quad (65)$$

is a measure the width of main lobe. (It's similar to a weighted average of the radiation intensity over the sphere.)

A **smaller** pattern solid angle means a **more directional antenna**. At one extreme, an isotropic antenna has $F(\theta, \phi) = 1$ for all θ and ϕ . In this case,

$$\Omega_p = \iint_{4\pi} d\Omega = 4\pi, \quad (66)$$

which is the largest possible value.

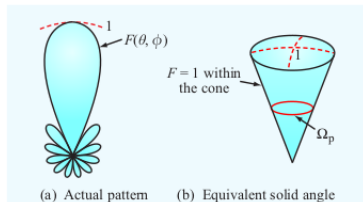


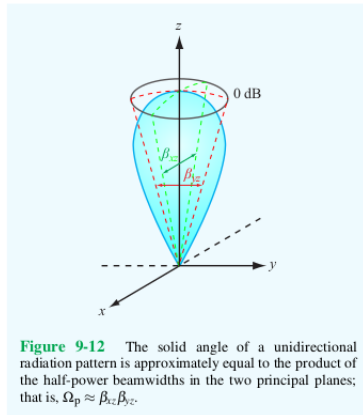
Figure 9-11 The pattern solid angle Ω_p defines an equivalent cone over which all the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.

Antenna Radiation Characteristics (Continued)

The main lobe of an antenna can be measured in a few ways. Each of these is called a **beamwidth**. These measures are as seen from the origin.

- **half-power beamwidth** β – angular width between points where the power in the main lobe is half its maximum
- **null beamwidth** β_{null} – angular width between first nulls on opposite sides of the peak

The half-power beamwidth is also called the **3 dB beamwidth** and the angles associated with this measure are the **half-power angles**.



Antenna Radiation Characteristics (Continued)

Note that

$$\Omega_p \approx \beta_{xz}\beta_{yz}, \quad (67)$$

where β_{xz} is the half-power beamwidth in the x - z plane and β_{yz} is the half-power beamwidth in the y - z plane.

Antenna Directivity

The **directivity**

$$D = \frac{F_{\max}}{F_{\text{av}}} \quad (68)$$

of an antenna is a measure of how directional the antenna is. Here, $F_{\max} = 1$ so that

$$D = \frac{1}{\frac{1}{4\pi} \iint_{4\pi} F(\theta, \phi) d\Omega} = \frac{4\pi}{\Omega_p}. \quad (69)$$

A larger value of D means that the energy is emitted over a smaller range of angles. Since the maximum value of Ω_p is 4π , the minimum value of D is 1 and this is the directivity of an isotropic antenna.

Antenna Radiation Characteristics (Continued)

An alternative expression for the directivity is

$$D = \frac{S_{\max}}{S_{\text{av}}}, \quad (70)$$

where $S_{\text{av}} = P_{\text{rad}}/(4\pi R^2)$. Also,

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \quad (71)$$

which is based on measures of the main lobe.

Antenna Gain

The **radiation efficiency** ξ of an antenna is

$$\xi = \frac{P_{\text{rad}}}{P_t}, \quad (72)$$

where P_t is the power that the generator applies to the antenna.

Antenna Radiation Characteristics (Continued)

Then the antenna **gain** is

$$G = \frac{4\pi R^2 S_{\max}}{P_t}. \quad (73)$$

Note that

$$G = \xi D. \quad (74)$$

Antenna Radiation Characteristics (Continued)

Then the antenna **gain** is

$$G = \frac{4\pi R^2 S_{\max}}{P_t}. \quad (75)$$

Note that

$$G = \xi D. \quad (76)$$

The **gain accounts for power dissipated** in the system whereas the directivity ignores these losses.

In the ideal case of a **lossless antenna**, $\xi = 1$ and $G = D$.

Radiation Resistance

An antenna is a load on the transmission line connecting the generator and the antenna with **input impedance**

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}}, \quad (77)$$

where the **resistive** component R_{in} corresponds to a resistor

Antenna Radiation Characteristics (Continued)

with total power P_t and current I_0 flowing through it. This means

$$P_t = \frac{1}{2} I_0^2 R_{\text{in}} \quad (78)$$

The total power can be split into a component P_{rad} that is **radiated** and a component P_{loss} that is **dissipated** so that

$$P_t = P_{\text{rad}} + P_{\text{loss}} \quad (79)$$

and the resistance R_{in} can be split into a **radiation resistance** R_{rad} and a **loss resistance** R_{loss} with

$$R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}, \quad (80)$$

where

$$P_{\text{rad}} = \frac{1}{2} I_0^2 R_{\text{rad}} \quad \text{and} \quad P_{\text{loss}} = \frac{1}{2} I_0^2 R_{\text{loss}}. \quad (81)$$

Antenna Radiation Characteristics (Continued)

The **radiation resistance** can be determined by inverting the above equation, i.e.

$$R_{\text{rad}} = \frac{2}{I_0^2} P_{\text{rad}}. \quad (82)$$

Also, the **radiation efficiency** can be written as

$$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}. \quad (83)$$

Homework

Read Chapter 9.

Homework 3 is posted and due Tuesday, May 26th,
at the beginning of class.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.
Notice that this is 10 minutes longer than our usual class time.
(Please verify this and let me know if I'm wrong.)